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From this assignment, I learned how to configure and apply **standard Access Control Lists (ACLs)** on Cisco routers to control network traffic. I understood that a standard ACL mainly filters traffic based on the **source IP address**. In this activity, I configured ACL 1 on R2 to block the 192.168.11.0/24 network from accessing the WebServer while allowing other traffic. I also configured ACL 1 on R3 to prevent the 192.168.10.0/24 network from communicating with the 192.168.30.0/24 network. I learned that an ACL must be applied to an interface using `ip access-group` before it can filter traffic. I also learned about the implicit deny rule and why `permit any` is needed when other traffic should be allowed. Finally, I used ping tests and show commands to verify that the ACLs were working correctly.

The image shows a network diagram and configuration steps for ACLs. The diagram illustrates a network topology with three routers (R1, R2, R3) and three PCs (PC1, PC2, PC3). R1 is connected to R2 and R3. R2 is connected to R3 and a WebServer. R3 is connected to R2 and a WebServer. PC1 is connected to R1, PC2 to R2, and PC3 to R3. The diagram also shows the configuration of ACL 1 on R2 and R3, and the application of the ACL to the GigabitEthernet0/0 interface on R3.

Step 2: Configure and apply a numbered standard ACL on R3.

- Create an ACL using the number 1 on **R3** with a statement that denies access to the 192.168.30.0/24 network from the **PC1** (192.168.10.0/24) network.
`R3(config)# access-list 1 deny 192.168.10.0 0.0.0.255`
- By default, an ACL denies all traffic that does not match a rule. To permit all other traffic, create a second rule for ACL 1.
`R3(config)# access-list 1 permit any`
- Apply the ACL by placing it for outbound traffic on the Gigabit Ethernet 0/0 interface.
`R3(config)# interface GigabitEthernet0/0`
`R3(config-if)# ip access-group 1 out`

Step 3: Verify ACL configuration and functionality.

- On **R2** and **R3**, enter the `show access-list` command to verify the ACL configurations. Enter the `show run` or `show ip interface gigabitethernet 0/0` command to verify the ACL placements.
- With the two ACLs in place, network traffic is restricted according to the policies detailed in Part 1. Use the following tests to verify the ACL implementations:
 - A ping from 192.168.10.10 to 192.168.11.10 succeeds.
 - A ping from 192.168.10.10 to 192.168.20.254 succeeds.
 - A ping from 192.168.11.10 to 192.168.20.254 fails.
 - A ping from 192.168.10.10 to 192.168.30.10 fails.
 - A ping from 192.168.11.10 to 192.168.30.10 succeeds.
 - A ping from 192.168.30.10 to 192.168.20.254 succeeds.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num
Successful	PC1	PC2	ICMP		0.000	N	0	
Successful	PC2	PC1	ICMP		0.000	N	1	